



Name:

Teacher:

Year 12
Mathematics
Trial HSC

August, 2017

Time allowed: 3 hours plus 5 minutes reading time

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A reference sheet is provided

Total marks – 100

Section I – 10 Marks

- Attempt Question 1-10 on the sheet provided
- Allow about 15 minutes for this section

Section II – 90 Marks

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. The solution to $x^2 - 4 < -3$ is:

- (A) $-2 < x < 2$
- (B) $x < -2, x > 2$
- (C) $-1 < x < 1$
- (D) $x < -1, x > 1$

2. An infinite geometric series has first term 4 and a limiting sum of 6.
What is the common ratio?

- (A) $\frac{1}{6}$
- (B) $\frac{1}{5}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{3}$

3. What is a possible primitive function for $2x^{-4} + 5x$?

- (A) $-\frac{2}{3x^3} + \frac{5x^2}{2} + 12$
- (B) $-\frac{1}{6x^3} + \frac{5x^2}{2}$
- (C) $\frac{2}{3x^3} + \frac{5x^2}{2} + 12$
- (D) $\frac{1}{6x^3} + \frac{5x^2}{2}$

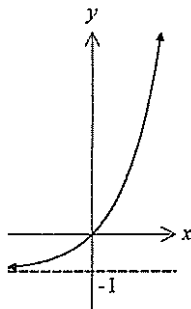
4. The quadratic equation $x^2 + 5x - 4 = 0$ has roots α and β .
What is the value of $2\alpha^2\beta + 2\alpha\beta^2$?

(A) -20
(B) 40
(C) -40
(D) 20

5. What are the solutions of $\tan 2\theta = 1$ for $0 \leq \theta \leq 360^\circ$?

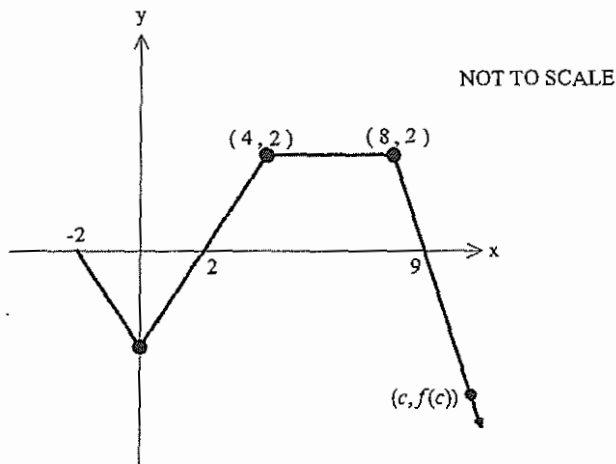
(A) $\theta = 45^\circ, 225^\circ$
(B) $\theta = 45^\circ, 225^\circ, 405^\circ, 585^\circ$
(C) $\theta = 22\frac{1}{2}^\circ, 112\frac{1}{2}^\circ$
(D) $\theta = 22\frac{1}{2}^\circ, 112\frac{1}{2}^\circ, 202\frac{1}{2}^\circ, 292\frac{1}{2}^\circ$

6. What is a possible equation for the following graph?



(A) $y = e^{x-1}$
(B) $y = e^x + 1$
(C) $y = e^x - 1$
(D) $y = e^{x+1}$

7. Consider the graph below:



For what value of C would $\int_{-2}^C f(x) dx = -2$ be true?

- (A) 10
 (B) 11
 (C) 12
 (D) 13
8. What is the value of $\sum_{n=1}^5 n(n-1)$?

- (A) 50
 (B) 40
 (C) 30
 (D) 20

9. For what values of x is the curve $f(x) = 2x^3 + x^2$ both concave down and decreasing?

(A) $-\frac{1}{6} < x < 0$

(B) $-3 < x < 0$

(C) $-3 < x < -\frac{2}{12}$

(D) $-\frac{1}{3} < x < -\frac{1}{6}$

10. A parabola has a focus $(0,6)$ and directrix of $y = 2$.

What is the equation of the parabola?

(A) $x^2 = -8(y - 4)$

(B) $x^2 = -16(y - 5)$

(C) $x^2 = 8(y - 4)$

(D) $x^2 = 16(y - 5)$

Section II

Total marks – 90

Attempt Questions 11–16

Allow about 2 hours 45 minutes for this section.

Begin each question on a NEW page.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Begin a NEW page.

- a) Evaluate $\frac{7.4^2 - e^2}{\sqrt{12} - \sqrt{2}}$ to 4 significant figures. 2
- b) Rationalise the denominator of $\frac{5\sqrt{2}}{2\sqrt{2} - 3}$. 2
- c) Fully factorise $x^6 - 27$. 2
- d) Solve the equation $|5 - x| = 3x$. 2
- e) If $\sin\theta = \frac{7}{10}$ and $\tan\theta < 0$, find the exact value of $\sec\theta$. 2
- f) Simplify $\lim_{x \rightarrow \infty} \frac{2x^3 - 4x^2 + 7}{x^3 + 3x + 1}$ 2
- g) Find the equation of the normal to the curve $y = 4e^{2(x-1)}$ at $x = 1$. 3

End of Question 11

Question 12 (15 marks) Begin a NEW page.

a) Differentiate the following with respect to x

i. $(3x^2 + 4)^5$ 2

ii. $x^2 \tan x$ 2

iii. $\frac{\sin x}{e^{-x}}$ 2

b) Find the area under the curve $y = |2x - 1|$ bounded by $x = -4$ and $x = 2$ 2

c) Sketch the curve $y = 4 \sin(2x) + 1$ between $-\pi \leq x \leq \pi$ 3
showing all important features (You DO NOT need to find x-intercepts).
(Make your graph at least a third of a page)

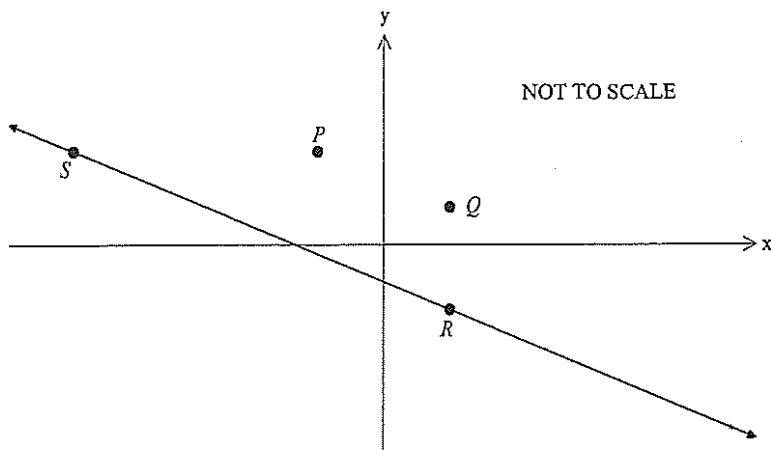
d) A function $y = f(x)$ has $\frac{dy}{dx} = 3x - 4$ and passes through $(1, 4)$. Find $f(x)$. 2

e) Shade the region represented by the intersection of $x^2 + (y - 3)^2 \leq 4$ 2
and $x + y > 3$.

End of Question 12

Question 13 (15 marks) Begin a NEW page.

- a) The points $P(-3,5)$ and $Q(3,2)$ are shown on the number plane below.



The equation of the line passing points S and R is $y = -\frac{1}{2}x - 2$

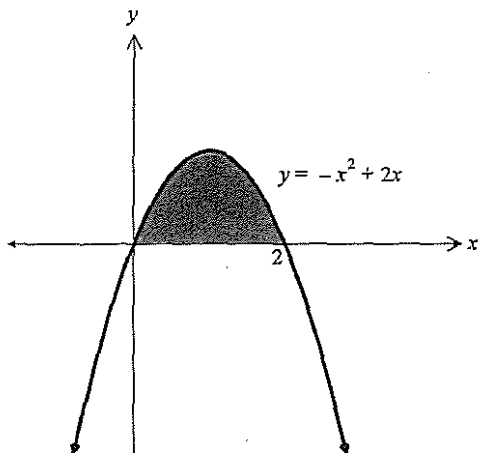
- | | | |
|------|--|---|
| i. | Find the gradient of PQ . Explain why PQRS is a trapezium. | 2 |
| ii. | Find the length of PQ in exact form. | 2 |
| iii. | Given that line QR is parallel to the y -axis, state the coordinates of R. | 1 |
| iv. | Find the perpendicular distance from P to the line RS . | 2 |
| v. | If the length of RS is $\sqrt{95}$ units find the area of PQRS correct to 2 decimal places | 2 |

Question 13 continues on page 9

Question 13 (continued)

- b) The graph of $y = -x^2 + 2x$ is shown below.

3



Find the volume of the solid of revolution formed when the shaded region is rotated about the x -axis.

- c) Given the function $f(x) = 3^{\cos x}$

- i. Copy and complete the table for $y = f(x)$ in your exam booklet.
(Round your answers to 3 decimal places)

1

x	0	1	2	3	4
y	3.000				

- ii. Apply the Trapezoidal rule with 4 subintervals to find an approximation of

2

$$\int_0^4 3^{\cos x} dx$$

correct to 2 decimal places.

End of Question 13

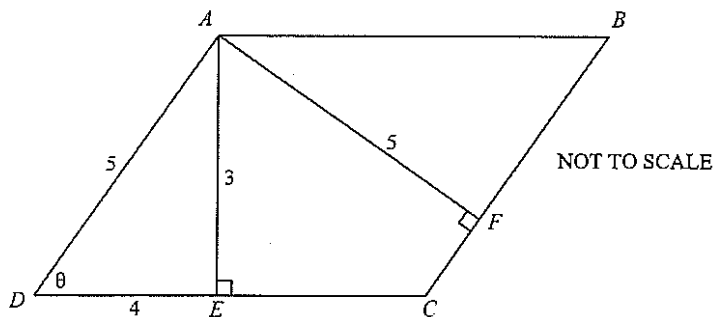
Question 14 (15 marks) Begin a NEW page.

- a) Consider the function

$$f(x) = \frac{x^2 + 15}{5x}$$

- | | | |
|------|--|---|
| i. | Show that the function is odd. | 1 |
| ii. | Show that there is no value of x for which $f(x) = 0$. | 1 |
| iii. | State the vertical asymptote of $y = f(x)$. | 1 |
| iv. | Find the stationary points and determine their nature. | 3 |
| v. | Sketch the graph of $y = f(x)$ showing all important features. | 2 |

- b) In the diagram below, $ABCD$ is a parallelogram.



Copy the diagram into your booklet

- | | | |
|-----|--|---|
| i. | Prove that if $\angle ADE = \theta$, then $\angle EAF = \theta$ (give reasons). | 2 |
| ii. | Hence, using the cosine rule, find the exact length of EF | 2 |
- c) Find the value of m for which the equation $(m - 4)x^2 - 6x + 7 = 0$ has one root twice the other.
- 3

End of Question 14

Question 15 (15 marks) Begin a NEW page.

a) Given the equation of the parabola $4y - 20 = x^2 + 12x + 36$:

- i. Find the coordinates of the vertex.
- ii. Find coordinates of the focus.
- iii. Find the equation of the directrix.

2

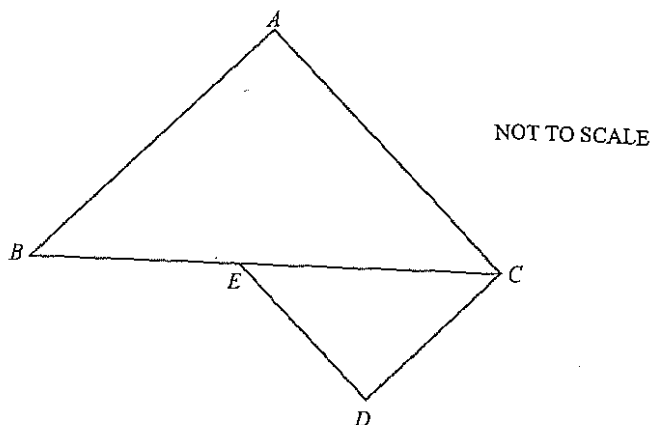
1

1

b) Find $\int (10x - 4)^5 dx$.

1

c) In the diagram $CD \parallel AB$ and $DE \parallel CA$. $AC = 15\text{cm}$, $AB = 18\text{cm}$, $CD = 8\text{cm}$ and $BE = 12\text{cm}$.



Copy the diagram into your booklet adding in all given information.

- i. Prove $\triangle ABC \parallel \triangle DCE$
- ii. Hence find the length of BC .

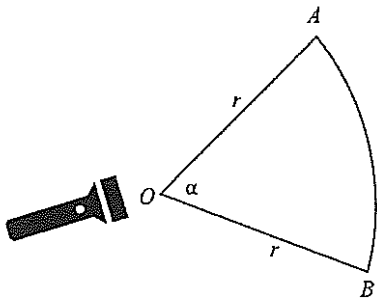
2

2

Question 15 continues on page 12

Question 15 (continued)

- d) The diagram below shows a sector OAB of a circle with centre O and a radius r cm created by the light of a torch.



- i. Show that the perimeter of the light sector OAB is $r(2 + \alpha)$ 1
- ii. Given that the perimeter of the light sector OAB is 6m, show that the area illuminated is given by: 2
- $$A = \frac{18\alpha}{(\alpha + 2)^2}$$
- iii. Hence show that the maximum illuminated area is 2.25m^2 . 3

End of Question 15

Question 16 (15 marks) Begin a NEW page.

a) Show that

3

$$\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = \frac{2}{\cos A}$$

b) The 4th term of an arithmetic sequence is 18 and the sum of the first 10 terms is 195.
Find the first term.

3

c) Mr Steve has a travel fund of \$55 000. The account accrues interest at 5.4% p.a. compounded monthly. He withdraws \$1 500 per month, after interest is paid, to pay for his travel adventures.

i. Show that the amount left at the end of the 2nd month is given by

2

$$A_2 = 55000 \times 1.0045^2 - 1500(1.0045 + 1)$$

ii. If A_n is the amount left after n months, show that:

2

$$A_n = 55000(1.0045)^n - 1500 \left[\frac{1.0045^n - 1}{0.0045} \right]$$

iii. Hence find the number of months Mr Steve can travel before his funds run out.

2

iv. If after 12 months Mr Steve decides to travel overseas, and increases his withdrawals to \$3 000 per month, how many more months can he now afford to travel.

3

End of Paper

Solutions
MULTIPLE CHOICE
ANSWER SHEET

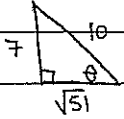


Mathematics 2 unit Trial HSC
August 2017

Completely fill the response oval representing the most correct answer.

Do not remove this sheet from the answer booklet.

1. A ☐ B ☐ C ☒ D ☐
2. A ☐ B ☐ C ☐ D ☒
3. A ☒ B ☐ C ☐ D ☐
4. A ☐ B ☒ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☒
6. A ☐ B ☐ C ☒ D ☐
7. A ☐ B ☐ C ☒ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☒
10. A ☐ B ☐ C ☒ D ☐

Q11	test: $ 5 - \frac{5}{4} = \frac{15}{4}$
a) $ 4.5596.... $	$3(\frac{5}{4}) = \frac{15}{4}$
≈ 14.56 (4 sig figs)	$\therefore \text{LHS} = \text{RHS}$
	$\therefore x = \frac{5}{4}$ only
b) $\frac{5\sqrt{2}}{2\sqrt{2}-3} \times \frac{2\sqrt{2}+3}{2\sqrt{2}+3}$	e) 
$= \frac{10 \times 2 + 15\sqrt{2}}{8-9}$	$\tan \theta < 0, \sin \theta > 0$ $\therefore \text{quadrant 2}$
$= \frac{20+15\sqrt{2}}{-1}$	$\sec \theta = -\frac{10}{\sqrt{51}}$ or $-\frac{10\sqrt{51}}{51}$
$= -20-15\sqrt{2}$	f. $\lim_{x \rightarrow \infty} \frac{2x^3 - 4x^2 + 7}{x^3 + 3x + 1} = 2$
c) $x^6 - 27$	g. $\frac{dy}{dx} = 8e^{2(x-1)}$
$= (x^2)^3 - 3^3$	at $x=1$
$= (x^2-3)(x^4+3x^2+9)$	$m_T = 8$; $m_N = -\frac{1}{8}$
d) $ 5-x = 3x$	Using $m = -\frac{1}{8}$ and $(1, 4)$
$5-x = -3x$	eqn of normal
$2x = -5$	$y-4 = -\frac{1}{8}(x-1)$
$x = -\frac{5}{2}$	$y-4 = -\frac{x}{8} + \frac{1}{8}$
test: $ 5 + \frac{5}{2} = \frac{15}{2}$	$y = -\frac{x}{8} + \frac{33}{8}$
$3(\frac{5}{2}) = -\frac{15}{2}$	$x+8y-33=0$
$\therefore \text{LHS} \neq \text{RHS}$	
$5-x = 3x$	
$5 = 4x$	
$x = \frac{5}{4}$	

Q12.

$$a) i. \frac{d}{dx} (3x^2+4)^5 = 5 \times 6x (3x^2+4)^4 \\ = 30x (3x^2+4)^4$$

$$ii. \frac{d}{dx} x^2 \tan x \Rightarrow$$

$$u = x^2 \quad v = \tan x$$

$$u' = 2x \quad v' = \sec^2 x$$

$$\frac{d}{dx} x^2 \tan x = 2x \tan x + x^2 \sec^2 x$$

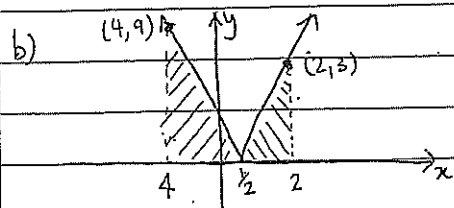
$$iii. \frac{d}{dx} \frac{\sin x}{e^{-x}}$$

$$u = \sin x \quad v = e^{-x}$$

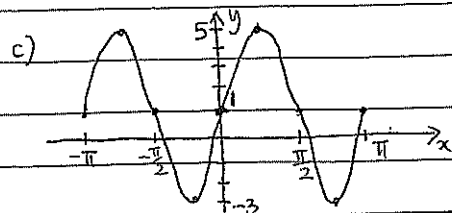
$$u' = \cos x \quad v' = -e^{-x}$$

$$\frac{d}{dx} \frac{\sin x}{e^{-x}} = \frac{\cos x e^{-x} + e^{-x} \sin x}{(e^{-x})^2}$$

$$= \frac{\cos x + \sin x}{e^{-x}}$$



$$\int_{-4}^2 |2x-1| dx = \frac{1}{2} \times 9 \times 9 + \frac{1}{2} \times 3 \times 3 \\ = 22\frac{1}{2} u^2$$



$$d) \frac{dy}{dx} = 3x-4$$

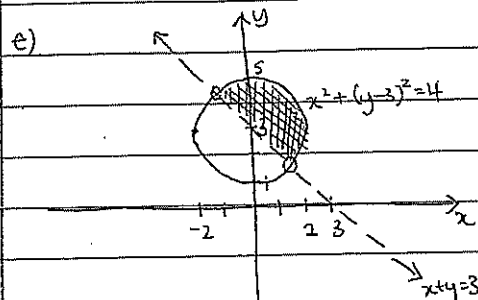
$$y = \int 3x-4 dx \\ = \frac{3}{2}x^2 - 4x + C$$

$$\text{sub } (1, 4)$$

$$4 = \frac{3}{2} - 4 + C$$

$$C = \frac{13}{2}$$

$$\therefore y = \frac{3}{2}x^2 - 4x + \frac{13}{2}$$



Q13.

$$a) i. m_{PQ} = \frac{5-2}{-3-3}$$

$$= \frac{3}{-6}$$

$$= -\frac{1}{2} = m_{RS}$$

$\therefore PQRS$ is a trapezium as $PQ \parallel RS$.

$$ii. d_{PR} = \sqrt{(-3-3)^2 + (5-2)^2}$$

$$= \sqrt{36+9}$$

$$= \sqrt{45}$$

$$= 3\sqrt{5}$$

$$iii. R(3, -\frac{7}{2})$$

$$iv. d_{PR} = \frac{|1(-3) + 2(5) + 4|}{\sqrt{1^2 + 2^2}}$$

$$= \frac{|11|}{\sqrt{5}}$$

$$= \frac{11\sqrt{5}}{5}$$

$$v. A = \frac{11\sqrt{5}}{5} \times \frac{1}{2} (\sqrt{95} + 3\sqrt{5})$$

$$= \frac{11\sqrt{5}}{10} (\sqrt{95} + 3\sqrt{5})$$

$$= 40.473...$$

$$\approx 40.47 u^2 \text{ (2 d.p.)}$$

$$b) V = \pi \int_0^2 (-x^2+2x)^2 dx$$

$$= \pi \int_0^2 x^4 - 4x^3 + 4x^2 dx$$

$$= \pi \left[\frac{x^5}{5} - x^4 + \frac{4}{3}x^3 \right]_0^2$$

$$= \pi \left(\frac{16}{15} \right)$$

$$= \frac{16\pi}{15} u^3$$

$$\approx 3.3510 u^3 \text{ (4 d.p.)}$$

c) i)

x	0	1	2	3	4
y	3.000	1.810	0.633	0.337	0.488

$$ii. \int_0^4 \cos x dx =$$

$$\approx \frac{1}{2} (3 + 0.488 + 2(1.810 + 0.633 + 0.337))$$

$$\approx 4.524$$

$$\approx 4.52 \text{ (2 d.p.)}$$

Q14.

$$\begin{aligned} \text{a) i. } f(-x) &= \frac{(-x)^2 + 15}{5(-x)} \\ &= \frac{x^2 + 15}{-5} \\ &= -\frac{x^2 + 15}{5} \\ &= -f(x) \end{aligned}$$

$\therefore$ odd function.

$$\begin{aligned} \text{ii. let } f(x) &= 0 \\ 0 &= \frac{x^2 + 15}{5x} \\ 0 &= x^2 + 15 \\ x^2 &= -15 \end{aligned}$$

no solution

$\therefore$ no x -value, $f(x) \neq 0$.

$$\text{iii. } x = 0$$

$$\begin{aligned} \text{iv. } y &= \frac{x^2 + 15}{5x} \\ &= \frac{x}{5} + \frac{3}{x} \end{aligned}$$

$$\frac{dy}{dx} = \frac{1}{5} - \frac{3}{x^2}$$

at $\frac{dy}{dx} = 0$, stationary pt.

$$0 = \frac{1}{5} - \frac{3}{x^2}$$

$$\frac{3}{x^2} = \frac{1}{5}$$

$$15 = x^2$$

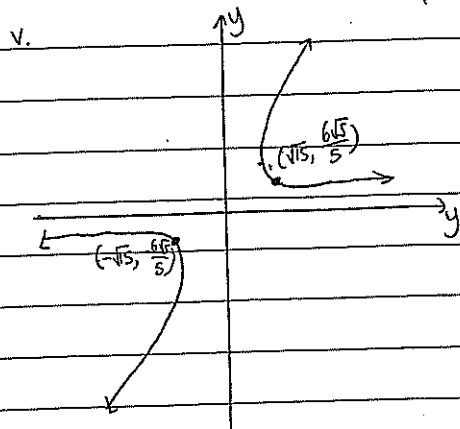
$$x = \pm\sqrt{15}$$

test:

x	-4	$-\sqrt{15}$	-3	3	$\sqrt{15}$	4
y'	$\frac{1}{80}$	0	$-\frac{2}{15}$	$-\frac{2}{15}$	0	$\frac{1}{80}$
	/	-	\	\	-	/

so: $(-\sqrt{15}, -\frac{6\sqrt{15}}{5})$ maximum

$(\sqrt{15}, \frac{6\sqrt{15}}{5})$ minimum



$$\text{b) i. } \angle ECF = 180^\circ - \theta^\circ$$

(co-interior angles, $AD \parallel BC$)

$$180^\circ - \theta^\circ + 90^\circ + 90^\circ + \angle EAF = 360^\circ$$

(angle sum of quadrilateral)

$$\angle EAF = \theta^\circ$$

$$\text{ii. } EF^2 = 3^2 + 5^2 - 2(3)(5)\cos\theta$$

$$\cos\theta = \frac{4}{5} \text{ (from } \triangle ADE)$$

$$EF^2 = 9 + 25 - 30 \times \left(\frac{4}{5}\right)$$

$$= 10$$

$$EF = \sqrt{10}$$

c) Roots are α and 2α

$$\text{so } \alpha + 2\alpha = -\frac{b}{a}$$

$$3\alpha = \frac{6}{m-4}$$

$$\alpha = \frac{2}{m-4}$$

$$\alpha \times 2\alpha = \frac{c}{a}$$

$$2\alpha^2 = \frac{7}{m-4}$$

$$\text{sub } \alpha = \frac{2}{m-4}$$

$$2\left(\frac{2}{m-4}\right)^2 = \frac{7}{m-4}$$

$$\frac{8}{(m-4)^2} = \frac{7}{m-4}$$

for quadratic, $m-4 \neq 0$ so divide off.

$$\frac{8}{m-4} = 7$$

$$8 = 7m - 28$$

$$7m = 36$$

$$m = \frac{36}{7} \text{ or } 5\frac{1}{7}$$

Q15	ii. $6 = r(2+\theta)$
a) $(x+6)^2 = 4(y-5)$	$r = \frac{6}{2+\theta}$
i. $V(-6, -5)$	Area of sector = $\frac{1}{2}r^2\theta$
ii $S(-6, -4)$	$= \frac{1}{2} \times \left(\frac{6}{2+\theta}\right)^2 \times \theta$
iii. $y = -6$	$= \frac{18\theta}{(2+\theta)^2}$
b) $\frac{(10x-4)^6}{60} + C$	iii. $A = \frac{18\theta}{(2+\theta)^2}$, $\theta \neq -2$
c) i. In $\triangle ABC$ and $\triangle DCE$	$u = 18\theta$ $v = (2+\theta)^2$
1. $\angle ACB = \angle DEC$	$u' = 18$ $v' = 2(2+\theta)$
(Alternate angles, $AC \parallel ED$)	$\frac{dA}{d\theta} = \frac{18(2+\theta)^2 - 36\theta(2+\theta)}{(2+\theta)^4}$
2. $\angle ABC = \angle ECD$	$= \frac{18(2+\theta) - 36\theta}{(2+\theta)^3}$
(Alternate angles, $AB \parallel CD$)	For max, $\frac{dA}{d\theta} = 0$
$\therefore \triangle ABC \parallel \triangle DCE$ (equiangular)	$0 = \frac{18(2+\theta) - 36\theta}{(2+\theta)^3}$
ii. $\frac{AB}{DC} = \frac{BC}{CE} = \frac{AC}{DE}$	$0 = 36 + 18\theta - 36\theta$
(matching sides in ratio, similar triangles)	$0 = 36 - 18\theta$
$\frac{18}{8} = \frac{12+CE}{CE}$	$18\theta = 36$
$18CE = 96 + 8CE$	$\theta = 2$
$10CE = 96$	test
$CE = 9.6$	$\theta \begin{array}{ c c c c } \hline & 1 & 2 & 3 \\ \hline A' & \frac{2}{3} & 0 & -\frac{18}{125} \\ \hline \end{array}$
$BE = 9.6 + 12$	$\therefore \text{max}$
$= 21.6 \text{ cm}$	Max Area = $\frac{18 \times 2}{(2+2)^2}$
d) i. $p = r + r + r\theta$ ($l = r\theta$)	$= 2.25 \text{ m}^2$
$= r(2+\theta)$	

Q16.	$= 55000 \times 1.0045^2 - 1500 \times 1.0045 - 1500$
a) LHS = $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A}$	$= 55000 \times 1.0045^2 - 1500(1.0045 + 1)$
$= \frac{\cos^2 A + (1+\sin A)^2}{\cos A(1+\sin A)}$	ii. $A_3 = 55000 \times 1.0045^3 - 1500(1.0045^2 + 1.0045 + 1)$
$= \frac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{\cos A(1+\sin A)}$	$A_n = 55000 \times 1.0045^n - 1500(1 + 1.0045 + \dots + 1.0045^{n-1})$
$= \frac{\cos^2 A + \sin^2 A + 1 + 2\sin A}{\cos A(1+\sin A)}$	$= 55000 \times 1.0045^n - 1500 \left[\frac{1(1.0045^n - 1)}{1.0045 - 1} \right]$
$= \frac{2 + 2\sin A}{\cos A(1+\sin A)}$	$= 55000 \times 1.0045^n - 1500 \left[\frac{1.0045^n - 1}{0.0045} \right]$
$= \frac{2(1+\sin A)}{\cos A(1+\sin A)}$	iii. $0 = 55000 \times 1.0045^n - 1500 \left(\frac{1.0045^n - 1}{0.0045} \right)$
$= \frac{2}{\cos A}$	$1500 \left(\frac{1.0045^n - 1}{0.0045} \right) = 55000 \times 1.0045^n$
$= \text{RHS}$	$1.0045^n - 1 = \frac{55000 \times 0.0045}{1500} \times 1.0045^n$
b) $T_4 = 18$	$1.0045^n = \frac{33}{200} (1.0045^n) = 1$
$18 = a + 3d$ ①	$\frac{167}{200} (1.0045^n) = 1$
$S_{10} = 195$	$1.0045^n = \frac{200}{167}$
$195 = 5(2a + 9d)$ ②	$\ln \frac{200}{167} = n$
$3 \times ① : 54 = 3a + 9d$	$\ln 1.0045$
$② : 39 = 2a + 9d$	$n = 40.16 \dots$
$15 = a + 0$	$\approx 40 \text{ months}$
$\therefore a = 15$	iv. $A_{12} = 55000 \times 1.0045^{12} - 1500 \left(\frac{1.0045^{12} - 1}{0.0045} \right)$
c) i. $A_1 = 55000 \times 1.0045 - 1500$	$= 39592.37071 \dots$
$A_2 = A_1 \times 1.0045 - 1500$	1 st month after 12 (i.e. 13 th)
$= (55000 \times 1.0045 - 1500) \times 1.0045 - 1500$	$A_1 = A_{12} \times 1.0045 - 3000$
	$A_2 = (A_{12} \times 1.0045 - 3000) \times 1.0045 - 3000$

$$= A_{12} \times 1.0045^2 - 3000(1 + 1.0045)$$

$$A_3 = A_{12} \times 1.0045^3 - 3000(1 + 1.0045 + 1.0045^2)$$

$$A_n = A_{12} \times 1.0045^n - 3000(1 + 1.0045 + \dots + 1.0045^{n-1})$$

$$= A_{12} \times 1.0045^n - 3000 \left[\frac{1.0045^n - 1}{0.0045} \right]$$

$$\text{Let } A_n = 0$$

$$0 = A_{12} \times 1.0045^n - 3000 \left[\frac{1.0045^n - 1}{0.0045} \right]$$

$$3000 \left[\frac{1.0045^n - 1}{0.0045} \right] = A_{12} \times 1.0045^n$$

$$\frac{1.0045^n - 1}{0.0045} = \frac{A_{12} \times 0.0045}{3000} \times 1.0045^n$$

$$1.0045^n - \left(\frac{A_{12} \times 0.0045}{3000} \right) \times 1.0045^n = 1$$

$$1.0045^n \left(1 - \frac{A_{12} \times 0.0045}{3000} \right) = 1$$

$$1.0045^n = 1 \div \left(1 - \frac{A_{12} \times 0.0045}{3000} \right)$$

$$\therefore = 1.068138245 \dots$$

$$n = \frac{\ln 1.068138245 \dots}{\ln 1.0045}$$

$$= 13.6361 \dots$$

$$\div 13$$

$\therefore$ He can travel 13 more months.

(can travel 25 months altogether)